

Hassan Ijaz

Huddersfield, UK | +44 7884527418 | oohhassanijaz@gmail.com | [LinkedIn profile](#) | [Portfolio](#) | [GitHub](#)

Personal Statement

Data Analyst with a strong academic background in Data Analytics and Computer Science, and proven experience delivering end-to-end machine learning solutions across structured datasets and real-world domains. Skilled in designing predictive models with a focus on accuracy, interpretability, and reproducibility. Proficient in advanced Python, SQL, and PostgreSQL, feature engineering, and building analytics-ready datasets for BI tools including Excel, Power BI and Tableau. Experienced in integrating diverse data sources, including macroeconomic and financial market data, to enable systematic analysis while ensuring strong data quality and statistical rigour.

Education & Qualifications

MSc DATA ANALYTICS | UNIVERSITY OF HUDDERSFIELD, UK | MAY 2025 – MAY 2026

- Modules are Machine Learning, Statistical Analysis, Big Data Analysis, Predictive Modelling, Python for Data Science, Data Visualization.

BSc COMPUTER SCIENCE | UNIVERSITY OF MANAGEMENT AND TECHNOLOGY, PAKISTAN | SEP 2019 – SEP 2023

- Final year project: *Bid Bazar*, a full-stack auction site, later acquired by my supervisor.

Certifications

AI-POWERED DATA ANALYTICS | CODEWITHHARRY, REMORT | NOV 2025 – APR 2026

- Advanced Python, SQL for analytics, advanced Excel, Git/GitHub, AI tools for analysts, probability and statistics, Power BI.

Key Project

MSc DISSERTATION | PREDICTING CO2 EMISSIONS FROM UK HOUSING DATA | NOV 2025 – MAY 2026 | [GITHUB LINK](#)

Technologies: Python, Pandas, NumPy, Scikit-Learn, XGBoost, Random Forest, Gradient Boosting, SHAP, Matplotlib.

Methods: Machine Learning, Predictive Modelling, Feature Engineering, Hyperparameter Tuning, Cross-Validation, Model Interpretability

- Architected an end-to-end predictive pipeline on 139,150 UK EPC residential records, applying IQR-based outlier removal to produce a refined dataset of 110,031 properties and engineering a structured feature framework from raw EPC description fields including roof insulation depth, wall and floor construction type, glazing properties, fuel group, and construction-year midpoint.
- Benchmarked four regression algorithms on an 80/20 train-test split, with XGBoost achieving R2 is 0.96 and MAE is 0.20 outperforming Gradient Boosting (R2 is 0.95) and Random Forest (R2 is 0.93), confirmed through 5-fold cross-validation and Grid Search hyperparameter tuning.
- Built a parallel High/Low emitter classification model on a median-split target, achieving 96.2% accuracy and F1 is 0.96, complementing the continuous regression output with a binary risk-banding view useful for retrofit prioritization policy.
- Applied SHAP-based interpretability analysis on the tuned XGBoost model to identify dominant emission drivers and covering exploratory analysis, model performance, and feature attribution insights.

Bid Bazar | Full-Stack Online Auction Platform | Sep 2022 – Sep 2023

Technologies: JavaScript, Node.js, Express.js, MongoDB, MVC Architecture, RESTful APIs.

Methods: Backend Development, Database Design, User Authentication, Real-Time Bidding Logic, Timer-Based Event Handling, Admin Dashboard Development.

- Spearheaded backend implementation of Bid Bazar, a full-featured online auction portal designing and building user login, bid placement, timer-enabled auction closure, and complete administrative functionality on a JavaScript and MongoDB stack with an MVC architecture.
- Engineered a MongoDB document data layer with collection schemas covering users, listings, bids, and transaction logs, ensuring data integrity and supporting concurrent auction flows with efficient query ability

across user and admin views.

- Built a session-based authentication and authorization system separating buyer, seller, and administrator roles, alongside admin-side controls for listing approval, user management, and live auction monitoring.
- Delivered a timer-driven auction closure engine with automatic winner determination and concurrent bid handling logic the platform saw heavy user engagement during deployment and was subsequently acquired by a furniture auction company for continued commercial operation.

Skills & Achievements

- **Programming:** Python, SQL, Git, GitHub
- **Data Analysis and ML:** Pandas, NumPy, Scikit-learn, XGBoost, Random Forest, Gradient Boosting, GridSearchCV, Cross-Validation, SHAP, Feature Engineering, Time Series Analysis
- **Visualization and BI:** Excel (Advanced), Power BI, Tableau, Matplotlib, Plotly, Interactive Dashboards, KPI Tracking and Reporting
- **Data Management:** MySQL, PostgreSQL, Database Design, ETL Pipelines, Data Cleaning, IQR Outlier Removal
- **Cloud and Data Platforms (Familiar):** Microsoft Azure, Google Big Query
- **Statistical Methods:** Hypothesis Testing, Predictive Modelling, Regression Analysis, Root-Cause Analysis
- **Financial Analysis:** Technical and fundamental analysis, equities earnings, macro data (CPI, GDP, rates, central bank policy), COT positioning, market microstructure, Risk Management, Position Sizing, Time Series Pattern Recognition
- **Web Development:** HTML, CSS, JavaScript, React.js, Laravel (PHP), RESTful APIs, Inertia.js
- **Business and Communication:** Stakeholder Management, Stakeholder Reporting, Data-Driven Documentation, Presentation of Insights, Requirements Gathering
- **Leadership:** Project Planning, Milestone Tracking, Cross-Functional Collaboration, Agile Working

Experience History

WEB DEVELOPER | AKVON IT SOLUTIONS, PAKISTAN | JUL 2023 – APR 2025

- Maintained client-facing web apps in React, Laravel/PHP, MySQL and Inertia.js as part of a small dev team.
- Built a session-based auth system with role-based access for production accounts.
- Optimised the schema across 11 relational databases and rewrote slow SQL on high-traffic endpoints.
- Developed 7 REST APIs and supporting pipelines, cutting data transfer time by 30%.
- Owned database design decisions and stakeholder communication for an Australian client.

LINKS AVAILABLE ON REQUEST, SUBJECT TO COMPANY RESTRICTIONS.

FINANCIAL MARKET ANALYST | 2021 – PRESENT

- Analysed financial markets independently since 2021 across crypto, forex, futures, options and equities covering both technical and fundamental sides.
- Read price action, chart patterns, volume and open interest across multiple timeframes.
- Studied multiple years of historical setups to understand how technical patterns typically work.
- Reviewed company earnings reports revenue, margins, guidance, balance sheets to build views on individual stocks.
- Tracked CPI, GDP, unemployment, central bank policy, interest rate decisions, and global news. Also, read COT reports to gauge institutional positioning.
- Produced monthly probability-weighted outlooks combining macro data, positioning, and technical setups.

Social Work

COMMUNITY VOLUNTEER | UNIVERSITY-AFFILIATED NGO, PAKISTAN | 2022 – 2023

- Led food drives and Ramadan iftar distribution for underprivileged families, coordinating donations and a team of volunteers.
- Worked across donors, volunteers and families from different backgrounds and picked up strong coordination, planning and people skills.